Chapter 4

Set Operators

Set operators are operators which are applicable to sets only. Five set operators will be studied in this chapter. They are: *union, intersection, relative complement, absolute complement*, and *Cartesian product*.

As with the logical operators, we may use the set operators to form *set expression*. Owing to the fact that most of the set expressions in the remaining part of this text are rather short, we shall make no attempt to give the concept a formal definition (the reader can easily make up one by modifying the definition of *well-formed formula*). Using set expressions as *terms* (see Section 2.10), and treating $\in$, $\subseteq$ and $=$ as predicates, we can form *FOL-wffs* involving set notations.

For a similar reason, we do not intend to provide an elaborated priority scheme for the set operators other than the following one: the unary operators have higher priority than the binary operators; unary operators are right associative while binary operators are left associative. Note that as set expressions are terms, set operators have higher priority than the logical operators and the predicates $\in$, $\subseteq$ and $=$.

4.1 Union

Definition: Let A, B be two sets. The ***union*** of A and B is the set $A \cup B = \{x \mid x \in A \vee x \in B\}$.

Example 4.1.1

(i) Let $A = \{a, b, c\}$ and $B = \{1, 2, 3\}$. Then $A \cup B = \{a, b, c, 1, 2, 3\}$.

(ii) Let $A = \{a, b, c, g, z\}$ and $B = \{a, c, r, x, y, z\}$. Then $A \cup B = \{a, b, c, g, r, x, y, z\}$.

The following are some basic properties of the union operator.

Theorem 4.1.1: Let A, B, C be sets.

(i) $A \cup \emptyset = A$.

(ii) $A \cup A = A$.

(iii) $A \cup B = B \cup A$.

(iv) $(A \cup B) \cup C = A \cup (B \cup C)$.

(v) $A \subseteq B$ iff $A \cup B = B$.

Proof:

(i) (Bidirectional proof)

$x \in (A \cup \emptyset) \Leftrightarrow x \in A \vee x \in \emptyset$ (Definition of $\cup$)

$\Leftrightarrow x \in A \vee false$ (Lemma 3.5.7)

$\Leftrightarrow x \in A$ (E6)

$\therefore (\forall x)(x \in (A \cup \emptyset) \Leftrightarrow x \in A)$ (Gen)

Hence, $A \cup \emptyset = A$ (Principle of Extension)

(ii) (Bidirectional proof)

$x \in A \cup A \Leftrightarrow x \in A \vee x \in A$ (Definition of $\cup$)

$\Leftrightarrow x \in A$ (E4)

$\therefore (\forall x)(x \in (A \cup A) \Leftrightarrow x \in A)$ (Gen)

Hence, $A \cup A = A$ (Principle of Extension)

(iii) (Bidirectional proof)

$x \in A \cup B \Leftrightarrow x \in A \vee x \in B$ (Definition of $\cup$)

$\Leftrightarrow x \in B \vee x \in A$ (E10)

$\Leftrightarrow x \in B \cup A$ (Definition of $\cup$)

$\therefore (\forall x)(x \in (A \cup B) \Leftrightarrow x \in (B \cup A))$ (Gen)

Hence, $A \cup B = B \cup A$ (Principle of Extension)

(iv) (Bidirectional proof)

$x \in ((A \cup B) \cup C) \Leftrightarrow x \in (A \cup B) \vee x \in C$ (Definition of $\cup$)

$\Leftrightarrow (x \in A \vee x \in B) \vee x \in C$ (Definition of $\cup$)

$\Leftrightarrow x \in A \vee (x \in B \vee x \in C)$ (E12)

$\Leftrightarrow x \in A \vee x \in (B \cup C)$ (Definition of $\cup$)

$\Leftrightarrow x \in (A \cup (B \cup C))$ (Definition of $\cup$)

$\therefore (\forall x)(x \in ((A \cup B) \cup C) \Leftrightarrow x \in (A \cup (B \cup C)))$ (Gen)

Hence, $(A \cup B) \cup C = A \cup (B \cup C)$ (Principle of Extension)

(v) (Bidirectional proof)

$\Rightarrow$) (Direct proof) Suppose $A \subseteq B$.

We are to prove that $(A \cup B) = B$. For this we shall prove that $(\forall x)(x \in (A \cup B) \Leftrightarrow x \in B)$ or equivalently, $(\forall x)((x \in A \vee x \in B) \Leftrightarrow x \in B)$

(Bidirectional proof)

$\Rightarrow$) (Proof by cases)

Case (i): To prove that $x \in A \Rightarrow x \in B$.

Proof: $A \subseteq B$ Hypothesis

$\Rightarrow (\forall x)(x \in A \Rightarrow x \in B)$ (Definition of $\subseteq$)

$\Rightarrow (x \in A \Rightarrow x \in B)$ (UI)

Case (ii): To prove that $x \in B \Rightarrow x \in B$.

Proof: By Lemma 3.2.3(i), $B \subseteq B$.

Therefore, $(\forall x)(x \in B \Rightarrow x \in B)$. (Definition of $\subseteq$)

$\Rightarrow (x \in B \Rightarrow x \in B)$ (UI)

Hence, $(x \in A \vee x \in B) \Rightarrow x \in B$.

It follows that $x \in (A \cup B) \Rightarrow x \in B$. (Definition of $\cup$)

$\Leftarrow$) (Direct proof) $x \in B \Rightarrow x \in B \vee x \in A$ (I1)

$\Rightarrow x \in A \vee x \in B$ (E10)

$\Rightarrow x \in (A \cup B)$. (Definition of $\cup$)

We thus have: $x \in (A \cup B) \Leftrightarrow x \in B$

$\Rightarrow (\forall x)(x \in (A \cup B) \Leftrightarrow x \in B)$ (Gen)

$\Rightarrow (A \cup B) = B$. (Princple of Extension)

$\Leftarrow$) (Direct proof) Suppose $A \cup B = B$. . . . (I)

We are to prove that $A \subseteq B$. We shall prove that $(x \in A \Rightarrow x \in B)$.

(Direct proof) $x \in A \Rightarrow x \in A \vee x \in B$ (I1)

$\Rightarrow x \in (A \cup B)$ (Definition of $\cup$)

$\Rightarrow x \in B$ ((I), $\text{Sub}_=$)

Hence, $x \in A \Rightarrow x \in B$.

$\Rightarrow (\forall x)(x \in A \Rightarrow x \in B)$ (Gen)

$\Rightarrow A \subseteq B$. (Definition of $\subseteq$)

In conclusion, we have: $A \subseteq B$ iff $A \cup B = B$. ■

4.2 Intersection

Definition: Let A, B be two sets. The ***intersection*** of A and B is the set $A \cap B = \{x \mid x \in A \wedge x \in B\}$.

Example 4.2.2

(i) Let $A = \{a, b, c\}$ and $B = \{1, 2, 3\}$. Then $A \cap B = \emptyset$.

(ii) Let $A = \{a, b, c, g, z\}$ and $B = \{a, c, r, x, y, z\}$. Then $A \cap B = \{a, c, z\}$.

The following are some basic properties of the intersection operator.

Theorem 4.2.2: Let A, B, C be sets.

(i) $A \cap \emptyset = \emptyset$.

(ii) $A \cap A = A$.

(iii) $A \cap B = B \cap A$.

(iv) $(A \cap B) \cap C = A \cap (B \cap C)$.

(v) $A \subseteq B$ iff $A \cap B = A$.

Proof:

(i) (Bidirectional proof)

$x \in (A \cap \emptyset) \Leftrightarrow x \in A \wedge x \in \emptyset$ $(Definition\ of \cap)$

$\Leftrightarrow x \in A \wedge false$ (Lemma 3.5.7)

$\Leftrightarrow false$ (E7)

$\Leftrightarrow x \in \emptyset$ (Lemma 3.5.7)

$\therefore (\forall x)(x \in A \cap \emptyset \Leftrightarrow x \in \emptyset)$. (Gen)

Hence, $A \cap \emptyset = \emptyset$. (Principle of Extension)

(ii) (Bidirectional proof)

$x \in A \cap A \Leftrightarrow x \in A \wedge x \in A$ (Definition of $\cap$)

$\Leftrightarrow x \in A$ (E3)

$\therefore (\forall x)(x \in A \Leftrightarrow x \in A)$. (Gen)

Hence, $A \cap A = A$. (Principle of Extension)

(iii) (Bidirectional proof)

$x \in A \cap B \Leftrightarrow x \in A \wedge x \in B$ (Definition of $\cap$)

$\Leftrightarrow x \in B \wedge x \in A$ (E9)

$\Leftrightarrow x \in B \cap A$ (Definition of $\cap$)

$\therefore (\forall x)(x \in A \cap B \Leftrightarrow x \in B \cap A)$. (Gen)

Hence, $A \cap B = B \cap A$. (Principle of Extension)

(iv) (Bidirectional proof)

$x \in ((A \cap B) \cap C) \Leftrightarrow x \in (A \cap B) \wedge x \in C$ (Definition of $\cap$)

$\Leftrightarrow (x \in A \wedge x \in B) \wedge x \in C$ (Definition of $\cap$)

$\Leftrightarrow x \in A \wedge (x \in B \wedge x \in C)$ (E11)

$\Leftrightarrow x \in A \wedge x \in (B \cap C)$ (Definition of $\cap$)

$\Leftrightarrow x \in (A \cap (B \cap C))$ (Definition of $\cap$)

$\therefore (\forall x)(x \in (A \cap B) \cap C \Leftrightarrow x \in A \cap (B \cap C))$. (Gen)

Hence, $(A \cap B) \cap C = A \cap (B \cap C)$. (Principle of Extension)

(v) (Bidirectional proof)

$\Rightarrow$) (Direct proof) Suppose $A \subseteq B$.

We are to prove that $(A \cap B) = A$, or equivalently, $(\forall x)(x \in (A \cap B) \Leftrightarrow x \in A)$.

We shall first prove that $x \in (A \cap B) \Leftrightarrow x \in A$.

(Bidirectional proof)

$\Rightarrow$) (Direct proof) $x \in (A \cap B) \Rightarrow x \in A \wedge x \in B$ (Definition of $\cap$)

$\Rightarrow x \in A$ (I2)

$\therefore x \in (A \cap B) \Rightarrow x \in A$.

$\Leftarrow$) (Direct proof) Suppose $x \in A$. . . . (I)

Since $A \subseteq B$, we have: $(\forall x)(x \in A \Rightarrow x \in B)$ (Definition of $\subseteq$)

$\Rightarrow (x \in A \Rightarrow x \in B)$. (UI) . . . (II)

$\Rightarrow x \in B$ ((I), (II), I3) . . . (III)

$\Rightarrow (x \in A \wedge x \in B)$ ((I),(III),I6)

$\Rightarrow x \in (A \cap B)$ (Definition of $\cap$)

Therefore, $x \in A \Rightarrow x \in (A \cap B)$

We thus have $(x \in (A \cap B) \Leftrightarrow x \in A)$

$\Rightarrow (\forall x)(x \in (A \cap B) \Leftrightarrow x \in A)$ (Gen)

$\Rightarrow (A \cap B) = A$. (Principle of Extension)

Hence, if $A \subseteq B$ then $(A \cap B) = A$.

$\Leftarrow$) (Direct proof) Suppose $(A \cap B) = A$. . . . (IV)

We are to prove that $A \subseteq B$, or equivalently, $(\forall x)(x \in A \Rightarrow x \in B)$.

First, we prove that $x \in A \Rightarrow x \in B$.

(Direct proof) Let $x \in A$.

Then $x \in (A \cap B)$ ((IV), sub$_=$)

$\Rightarrow x \in A \wedge x \in B$ (Definition of $\cap$)

$\Rightarrow x \in B \wedge x \in A$ (E9)

$\Rightarrow x \in B$ (I2)

We thus have $x \in A \Rightarrow x \in B$

$\Rightarrow (\forall x)(x \in A \Rightarrow x \in B)$ (Gen)

$\Rightarrow A \subseteq B$. (Definition of $\subseteq$)

Hence, if $(A \cap B) = A$, then $A \subseteq B$.

Thus $A \subseteq B$ iff $A \cap B = A$. ■

Corollary 4.2.2.1 Let A, B be any two sets. Then, $A \cap B \subseteq A \subseteq A \cup B$.

Proof: First, we want to prove that $A \cap B \subseteq A$ or equivalently $(\forall x)(x \in A \cap B \Rightarrow x \in A)$.

(Direct proof)

$x \in (A \cap B) \Rightarrow x \in A \wedge x \in B$ (Definition of $\cap$)

$\Rightarrow x \in A$ (I2)

We thus have $x \in (A \cap B) \Rightarrow x \in A$.

$\Rightarrow (\forall x)(x \in (A \cap B) \Rightarrow x \in A)$. (Gen)

$\Rightarrow A \cap B \subseteq A$. (Definition of $\subseteq$)

Next, we want to prove that $A \subseteq A \cup B$ or equivalently $(\forall x)(x \in A \Rightarrow x \in A \cap B)$.

(Direct proof)

$x \in A \Rightarrow x \in A \vee x \in B$ (I1)

$\Rightarrow x \in A \cup B$ (Definition of $\cup$)

We thus have $x \in A \Rightarrow x \in (A \cup B)$.

$\Rightarrow (\forall x)(x \in A \Rightarrow x \in (A \cup B))$. (Gen)

$\Rightarrow A \subseteq A \cup B$. (Definition of $\subseteq$) ■

Theorem 4.2.3: Let A, B, C be any three sets.

(i) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

(ii) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Proof: (Bidirectional proof)

(i) $x \in A \cup (B \cap C) \Leftrightarrow x \in A \vee x \in (B \cap C)$ (Definition of $\cup$)

$\Leftrightarrow x \in A \vee (x \in B \wedge x \in C)$ (Definition of $\cap$)

$\Leftrightarrow (x \in A \vee x \in B) \wedge (x \in A \vee x \in C)$ (E14)

$\Leftrightarrow x \in (A \cup B) \wedge x \in (A \cup C)$ (Definition of $\cup$)

$\Leftrightarrow x \in (A \cup B) \cap (A \cup C)$ (Definition of $\cap$)

$\therefore (\forall x)(x \in A \cup (B \cap C) \Leftrightarrow x \in (A \cup B) \cap (A \cup C))$ (Gen)

Hence, $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ (Principle of Extension)

(ii) (Similar to Part (i)) ■

Definition: Two sets A and B are ***disjoint*** if $A \cap B = \emptyset$.

4.3 Complements

Definition: Let A, B be two sets. The ***relative complement*** of B in A is the set $A - B = \{x \mid x \in A \wedge x \notin B\}$.

Example 4.3.3

(i) Let $A = \{1, 2, 3, 4, 5\}$, $B = \{1, 3, 5, 7, 9\}$, $C = \{1, 2, 3\}$ and $D = \{6, 7, 8\}$. Then $A - B = \{2, 4\}$, $B - A = \{7, 9\}$, $C - A = \emptyset$, $D - A = D$, and $A - D = A$.

(ii) Prove that $A - B \neq B - A$. (i.e. the relative complement is non-commutative)

(**Note:** to prove $A - B \neq B - A$ is to disprove $A - B = B - A$.)

Proof: (Proof by counterexample)

Let $A = \{a\}$ and $B = \emptyset$.

Then $A - B = \{a\} - \emptyset = \{a\}$ whereas $B - A = \emptyset - \{a\} = \emptyset$. . . . (I)

Since $a = a$ (Axiom A1 of equality)

Therefore, $a \in \{a\}$ (Definition of $\{a\}$)

$\Rightarrow (\exists x)(x \in \{a\})$ (EQ)

$\Rightarrow \{a\} \neq \emptyset$ (Lemma 3.5.8).

Therefore, $A - B \neq B - A$. ((I), Sub$_=$) ■

(iii) Prove that $(A - B) - C \neq A - (B - C)$. (i.e. the relative complement is non-associative)

Proof: (Proof by counterexample)

Let $A = B = C = \{a\}$

Then $(A - B) - C = (\{a\} - \{a\}) - \{a\} = \emptyset - \{a\} = \emptyset$,

whereas $A - (B - C) = \{a\} - (\{a\} - \{a\}) = \{a\} - \emptyset = \{a\}$.

Since $\{a\} \neq \emptyset$ (see Part (ii)). Therefore, $(A - B) - C \neq A - (B - C)$. ■

The following are some basic properties of the relative complement operator.

Theorem 4.3.4: Let A, B, C be sets.

(i) $A \subseteq B$ iff $A - B = \emptyset$.
(ii) $A - (A - B) = A \cap B$.
(iii) $A \cap (B - C) = (A \cap B) - (A \cap C)$.

Proof:

(i) (Bidirectional proof)

$A - B = \emptyset \Leftrightarrow \sim (\exists x) x \in (A - B)$ (Lemma 3.5.8 and Theorem 1.7.2(iii))

$\Leftrightarrow \sim (\exists x)(x \in A \wedge x \notin B)$ (Definition of $-$)

$\Leftrightarrow \sim (\exists x)(\sim\sim x \in A \wedge x \notin B)$ (E15)

$\Leftrightarrow \sim (\exists x) \sim (\sim x \in A \vee x \in B)$ (E17)

$\Leftrightarrow \sim (\exists x) \sim (x \in A \Rightarrow x \in B)$ (E18)

$\Leftrightarrow \sim\sim (\forall x)(x \in A \Rightarrow x \in B)$ (FE7)

$\Leftrightarrow (\forall x)(x \in A \Rightarrow x \in B)$ (E15)

$\Leftrightarrow A \subseteq B$ (Definition of $\subseteq$)

Hence, $A - B = \emptyset$ iff $A \subseteq B$.

(ii) (Bidirectional proof)

$x \in (A - (A - B)) \Leftrightarrow x \in A \wedge x \notin (A - B)$ (Definition of $-$)

$\Leftrightarrow x \in A \wedge \sim (x \in A \wedge x \notin B)$ (Definition of $-$)

$\Leftrightarrow x \in A \wedge (x \notin A \vee \sim x \notin B)$ (E16)

$\Leftrightarrow x \in A \wedge (x \notin A \vee x \in B)$ (E15)

$\Leftrightarrow (x \in A \wedge x \notin A) \vee (x \in A \wedge x \in B)$ (E13)

$\Leftrightarrow false \vee (x \in A \wedge x \in B)$ (E1)

$\Leftrightarrow (x \in A \wedge x \in B) \vee false$ (E10)

$\Leftrightarrow (x \in A \wedge x \in B)$ (E6)

$\Leftrightarrow x \in A \cap B$ (Definition of $\cap$)

$\therefore (\forall x)(x \in (A - (A - B)) \Leftrightarrow x \in (A \cap B))$ (Gen)

Hence, $A - (A - B) = A \cap B$ (Principle of Extension)

(iii) (Bidirectional proof)

$x \in (A \cap (B - C)) \Leftrightarrow x \in A \wedge x \in (B - C)$ (Definition of $\cap$)

$\Leftrightarrow x \in A \wedge (x \in B \wedge x \notin C)$ (Definition of $-$)

$\Leftrightarrow (x \in A \wedge (x \in B \wedge x \notin C)) \vee false$ (E6)

$\Leftrightarrow false \vee (x \in A \wedge (x \in B \wedge x \notin C))$ (E10)

$\Leftrightarrow (x \in B \wedge false) \vee (x \in A \wedge (x \in B \wedge x \notin C))$ (E7)

$\Leftrightarrow (x \in B \wedge (x \in A \wedge x \notin A)) \vee (x \in A \wedge (x \in B \wedge x \notin C))$ (E1)

$\Leftrightarrow ((x \in A \wedge x \in B) \wedge x \notin A) \vee ((x \in A \wedge x \in B) \wedge x \notin C)$ (E9, E10)

$\Leftrightarrow (x \in A \wedge x \in B) \wedge (x \notin A \vee x \notin C)$ (E13)

$\Leftrightarrow (x \in A \wedge x \in B) \wedge \sim (x \in A \wedge x \in C)$ (E16)

$\Leftrightarrow x \in (A \cap B) \wedge \sim (x \in A \cap C)$ (Definition of $\cap$)

$\Leftrightarrow x \in (A \cap B) - (A \cap C)$ (Definition of $-$)

$\therefore (\forall x)(x \in (A \cap (B - C)) \Leftrightarrow x \in (A \cap B) - (A \cap C))$ (Gen)

Hence, $A \cap (B - C) = (A \cap B) - (A \cap C)$ (Principle of Extension) ■

Definition: The ***(relative) Universal set***, denoted by $\mathbf{U}$, is a set which contains every object (thereby includes every set) in our discussion. Let A be any set such that $A \subseteq \mathbf{U}$. The ***absolute complement*** of A is the set $\overline{A} = \mathbf{U} - A$.

Example 4.3.4 In *Elementary Number Theory*, the universal set **U** is the set of all positive integers. Let A be the set of all the prime numbers and B be the set of all positive even integers. Then $A \subseteq \mathbf{U}$ and $B \subseteq \mathbf{U}$. $\overline{A}$ is the set of all composite integers including the integer 1, and $\overline{B}$ is the set of positive odd integers. ■

Lemma 4.3.5: Let $A \subseteq \mathbf{U}$. Then, $x \in \overline{A} \Leftrightarrow x \notin A$.

Proof: (Bidirectional proof)

$\Rightarrow$) (Indirect proof) Suppose $x \in A$.

Then $x \in A \vee x \notin \mathbf{U}$	(I1)
$\Rightarrow x \notin \mathbf{U} \vee x \in A$	(E10)
$\Rightarrow x \notin \mathbf{U} \vee \sim x \notin A$	(E15)
$\Rightarrow \sim (x \in \mathbf{U} \wedge x \notin A)$	(E16)
$\Rightarrow \sim (x \in (\mathbf{U} - A))$	(Definition of $-$)
$\Rightarrow \sim x \in \overline{A}$	(Definition of $\overline{A}$)
$\Rightarrow x \notin \overline{A}$	(Definition of $\notin$)

$\Leftarrow$) (Direct proof) Suppose $x \notin A$.

Since $x \in \mathbf{U}$	(Definition of **U**)
We have $x \in \mathbf{U} \wedge x \notin A$	(I6)
$\Rightarrow x \in (\mathbf{U} - A)$	(Definition of $-$)
$\Rightarrow x \in \overline{A}$	(Definition of $\overline{A}$)

Hence, $x \in \overline{A} \Leftrightarrow x \notin A$. ■

Theorem 4.3.6: (DeMorgan's Theorem) Let A, B be any two subsets of **U**.

(i) $\overline{A \cup B} = \overline{A} \cap \overline{B}$
(ii) $\overline{A \cap B} = \overline{A} \cup \overline{B}$.

Proof:

(i) First, we shall prove that $x \in \overline{A \cup B} \Leftrightarrow x \in \overline{A} \cap \overline{B}$.

(Bidirectional proof)

$x \in \overline{A \cup B} \Leftrightarrow x \notin A \cup B$	(Lemma 4.3.5)
$\Leftrightarrow \sim x \in A \cup B$	(Definition of $\notin$)
$\Leftrightarrow \sim (x \in A \vee x \in B)$	(Definition of $\cup$)
$\Leftrightarrow (\sim x \in A \wedge \sim x \in B)$	(E17)
$\Leftrightarrow (x \in \overline{A} \wedge x \in \overline{B})$	(Lemma 4.3.5)
$\Leftrightarrow x \in \overline{A} \cap \overline{B}$	(Definition of $\cap$)

Hence, $x \in \overline{A \cup B} \Leftrightarrow x \in \overline{A} \cap \overline{B}$

$\Rightarrow (\forall x)(x \in \overline{A \cup B} \Leftrightarrow x \in \overline{A} \cap \overline{B})$ (Gen)

Hence, $\overline{A \cup B} = \overline{A} \cap \overline{B}$ (Principle of Extension)

(ii) Similar to Part (i). ■

Theorem 4.3.7: Let A, B be any two subsets of $\mathbf{U}$.

(i) $\overline{\overline{A}} = A$
(ii) $\overline{\emptyset} = \mathbf{U}$ and $\overline{\mathbf{U}} = \emptyset$.
(iii) $A \cap \overline{A} = \emptyset$ and $A \cup \overline{A} = \mathbf{U}$.
(iv) $A \subseteq B$ iff $\overline{B} \subseteq \overline{A}$.
(v) $A - B = A \cap \overline{B}$.

Proof:

(i) (Bidirectional proof)

$x \in \overline{\overline{A}} \Leftrightarrow x \notin \overline{A}$ (Lemma 4.3.5, with $\overline{A}$ replacing A)

$\Leftrightarrow x \in A$ (Lemma 4.3.5 and E15)

$\therefore (\forall x)\left(x \in \overline{\overline{A}} \Leftrightarrow x \in A\right)$ (Gen)

Hence, $\overline{\overline{A}} = A$ (Principle of Extension)

(ii) (Bidirectional proof)

$x \in \overline{\emptyset} \Leftrightarrow x \in \mathbf{U} \wedge x \notin \emptyset$ (Definition of $\overline{\emptyset}$)

$\Leftrightarrow x \in \mathbf{U} \wedge \sim x \in \emptyset$ (Definition of $\notin$)

$\Leftrightarrow x \in \mathbf{U} \wedge \sim false$ (Lemma 3.5.7)

$\Leftrightarrow x \in \mathbf{U} \wedge true$ (Theorem 1.7.2(i))

$\Leftrightarrow x \in \mathbf{U}$ (E5)

$\therefore (\forall x)\left(x \in \overline{\emptyset} \Leftrightarrow x \in \mathbf{U}\right)$ (Gen)

$\Rightarrow \overline{\emptyset} = \mathbf{U}$ (Principle of Extension)

$x \in \overline{\mathbf{U}} \Leftrightarrow x \in \mathbf{U} \wedge x \notin \mathbf{U}$ (Definition of $\overline{\mathbf{U}}$)

$\Leftrightarrow false$ (E1)

$\Leftrightarrow x \in \emptyset$ (Lemma 3.5.7)

$\therefore (\forall x)\left(x \in \overline{\mathbf{U}} \Leftrightarrow x \in \emptyset\right)$ (Gen)

$\Rightarrow \overline{\mathbf{U}} = \emptyset$ (Principle of Extension)

(iii) First, we prove that $x \in A \cap \overline{A} \Leftrightarrow x \in \emptyset$

(Bidirectional proof)

$x \in A \cap \overline{A} \Leftrightarrow x \in A \wedge x \in \overline{A}$ (Definition of $\cap$)

$\Leftrightarrow x \in A \wedge x \notin A$ (Lemma 4.3.5)

$\Leftrightarrow false$ (E1)

$\Leftrightarrow x \in \emptyset$ (Lemma 3.5.7)

Heence, $x \in A \cap \overline{A} \Leftrightarrow x \in \emptyset$

$\Rightarrow (\forall x)\left(x \in A \cap \overline{A} \Leftrightarrow x \in \emptyset\right)$ (Gen)

$\Rightarrow A \cap \overline{A} = \emptyset$ (Principle of Extension)

To prove the second part, we first note that by the definition of $\mathbf{U}$, $A \cup \overline{A} \subseteq \mathbf{U}$.
It thus remains to prove that $\mathbf{U} \subseteq A \cup \overline{A}$ or equivalently, $x \in \mathbf{U} \Rightarrow x \in \left(A \cup \overline{A}\right)$.
We shall use the *trivial proof* method.

$x \in \left(A \cup \overline{A}\right) \Rightarrow x \in A \vee x \in \overline{A}$ (Definition of $\cup$)

$\Rightarrow x \in A \vee x \notin A$ (Lemma 4.3.5)

$\Rightarrow true$ (E2)

Hence, $x \in \mathbf{U} \Rightarrow x \in (A \cup \overline{A})$

$\Rightarrow (\forall x)(x \in \mathbf{U} \Rightarrow x \in (A \cup \overline{A}))$ (Gen)

$\Rightarrow \mathbf{U} \subseteq (A \cup \overline{A})$. (Definition of $\subseteq$)

Consequently, $A \cup \overline{A} \subseteq \mathbf{U}$ and $\mathbf{U} \subseteq (A \cup \overline{A})$

$\Rightarrow A \cup \overline{A} = \mathbf{U}$. (Theorem 3.2.3(iii))

(iv) (Bidirectional proof)

$A \subseteq B \Leftrightarrow (\forall x)(x \in A \Rightarrow x \in B)$ (Definition of $\subseteq$)

$\Leftrightarrow x \in A \Rightarrow x \in B$ ($\Rightarrow$)UI; $\Leftarrow$) Gen)

$\Leftrightarrow x \notin B \Rightarrow x \notin A$ (E19)

$\Leftrightarrow x \in \overline{B} \Rightarrow x \in \overline{A}$ (Lemma 4.3.5)

$\Leftrightarrow (\forall x)(x \in \overline{B} \Rightarrow x \in \overline{A})$ ($\Rightarrow$)Gen; $\Leftarrow$) UI)

$\Leftrightarrow \overline{B} \subseteq \overline{A}$ (Definition of $\subseteq$)

Hence, $A \subseteq B$ iff $\overline{B} \subseteq \overline{A}$.

(v) (Bidirectional proof)

$x \in A - B \Leftrightarrow x \in A \wedge x \notin B$ (Definition of $-$)

$\Leftrightarrow x \in A \wedge x \in \overline{B}$ (Lemma 4.3.5)

$\Leftrightarrow x \in A \cap \overline{B}$ (Definition of $\cap$)

$\therefore (\forall x)(x \in A - B \Leftrightarrow x \in A \cap \overline{B})$ (Gen)

Hence, $A - B = A \cap \overline{B}$. (Principle of Extension) ■

4.4 Ordered Pairs

Definition: The ***ordered pair*** of a and b is the set $(a, b) = \{\{a\}, \{a, b\}\} = \{x \mid x = \{a\} \vee x = \{a, b\}\}$. Element a is called the ***first co-ordinator*** and element b is called the ***second co-ordinator***.

The following is a fundamental property of ordered pairs.

Lemma 4.4.8: $(a, b) = (x, y)$ implies that $a = x$ and $b = y$.

Proof: (Direct Proof) Suppose $(a, b) = (x, y)$.

Two cases are to be considered separately.

Case (i): $a = b$

Then $(a, b) = (a, a) = \{\{a\}, \{a, a\}\} = \{\{a\}, \{a\}\} = \{\{a\}\}$. (Sub$_=$, Example ?(ii))

But $(a, b) = (x, y) = \{\{x\}, \{x, y\}\}$.

Therefore, $\{\{a\}\} = \{\{x\}, \{x, y\}\}$. (Theorem 3.2.1(ii),(iii)) . . . . (I)

Since $\{x\} \in \{\{x\}, \{x, y\}\}$, therefore $\{x\} \in \{\{a\}\}$. ($\because$ (I), Sub$_=$)

$\Rightarrow \{x\} = \{a\}$ (Definition of $\{\{a\}\}$) . . . . (II)

As $a \in \{a\}$, we have $a \in \{x\}$ ($\because$ (II), Sub$_=$)

$\Rightarrow a = x$. (Definition of $\{x\}$) . . . . (A)

Similarly, $\{x, y\} \in \{\{x\}, \{x, y\}\} \Rightarrow \{x, y\} \in \{\{a\}\}$ ($\because$ (I), Sub$_=$)

$\Rightarrow \{x, y\} = \{a\}$ (Definition of $\{\{a\}\}$) . . . . (III)

Since $y \in \{x, y\}$, therefore $y \in \{a\}$ ($\because$ (III), Sub$_=$)

$\Rightarrow y = a$ (Definition of $\{a\}$)

$\Rightarrow (y = a) \wedge (a = b)$ (Hypothesis of Case (i), I6)

$\Rightarrow b = y.$ (Transitivity and symmetricity of =) . . . . (B)

From (A) and (B), we have $a = x$ and $b = y$.

Case (ii): $a \neq b$

Then $(a, b) = (x, y) \Rightarrow \{\{a\}, \{a, b\}\} = \{\{x\}, \{x, y\}\}$. . . . . (IV)

It follows that $\{x\} \in \{\{x\}, \{x, y\}\} \Rightarrow \{x\} \in \{\{a\}, \{a, b\}\}$ ($\because$ (IV), Sub$_=$)

$\Rightarrow \{x\} = \{a\} \vee \{x\} = \{a, b\}.$ (Def. of $\{\{a\}, \{a, b\}\}$)

$\Rightarrow \{x\} = \{a, b\} \vee \{x\} = \{a\}.$ (E10)

$\Rightarrow (\{x\} \neq \{a, b\} \Rightarrow \{x\} = \{a\}).$ (E15,E18) $\cdots\cdots$ (V)

Suppose $\{x\} = \{a, b\}.$ $\cdots\cdots$ (VI)

Then, $a, b \in \{a, b\} \Rightarrow a, b \in \{x\}$ ($\because$ (VI), Sub$_=$)

$\Rightarrow a = x \; and \; b = x$ (Definition of $\{x\}$)

$\Rightarrow a = b$ (Sub$_=$)

$\Rightarrow (a = b) \wedge (a \neq b)$ (Hypothesis of Case (ii), I6)

$\Rightarrow false$ (i.e. a contradiction!) (E1)

Hence $\{x\} \neq \{a, b\}$ $\cdots\cdots$ (VII)

$\Rightarrow \{x\} = \{a\}.$ ((VII),(V),I3) $\cdots\cdots$ (VIII)

But then $a \in \{a\} \Rightarrow a \in \{x\}$ ($\because$ (VIII), Sub$_=$)

$\Rightarrow a = x.$ (Definition of $\{x\}$) $\cdots\cdots$ (C)

Again, $\{a, b\} \in \{\{a\}, \{a, b\}\} \Rightarrow \{a, b\} \in \{\{x\}, \{x, y\}\}$ ($\because$ (IV), Sub$_=$)

$\Rightarrow \{a, b\} = \{x\} \vee \{a, b\} = \{x, y\}.$ (Def. of $\{\{x\}, \{x, y\}\}$)

$\Rightarrow \{x\} = \{a, b\} \vee \{a, b\} = \{x, y\}.$ (Symmetricity of =)

$\Rightarrow (\{x\} \neq \{a, b\} \Rightarrow \{a, b\} = \{x, y\}).$ (E15,E18) $\cdots\cdots$ (IX)

$\Rightarrow \{a, b\} = \{x, y\}.$ ((VII),(IX),I3) $\cdots\cdots$ (X)

It follows that $b \in \{a, b\} \Rightarrow b \in \{x, y\}$ ((X), Sub$_=$)

$\Rightarrow b = x \vee b = y.$ (Definition of $\{x, y\}$)

$\Rightarrow (b \neq x \Rightarrow b = y).$ (E15,E18) $\cdots\cdots$ (XI)

If $b = x$, then $a = b$ ((C), Sub$_=$)

$\Rightarrow (a = b) \wedge (a \neq b)$ (Hypothesis of Case (ii), I6)

$\Rightarrow false$ (i.e. a contradiction!) (E1)

Hence, $b \neq x$ $\cdots\cdots$ (XII)

$\Rightarrow b = y.$ ((XII),(XI),I3) $\cdots\cdots$ (D)

From (C) and (D), we thus have $a = x$ and $b = y$. ■

Lemma 4.4.9: $a = c \wedge b = d \Rightarrow (a, b) = (c, d)$

Proof: An immediate consequence of the substitutivity of equality. ■

4.5 Cartesian Product

Definition: Let A, B be two sets. The ***Cartesian Product*** of A and B is the set

$$A \times B = \{x \mid (\exists a)(\exists b)(a \in A \wedge b \in B \wedge x = (a,b))\} = \{(a,b) \mid a \in A \wedge b \in B\}.$$

Example 4.5.5

1. Let $A = \{a, b, c\}$, $B = \{1, 2\}$. Then $A \times B = \{(a,1), (a,2), (b,1), (b,2), (c,1), (c,2)\}$.
2. Let $\mathbf{R}$ be the set of all real numbers. Then, $\mathbf{R}\times\mathbf{R} = \{(x,y) \mid x, y \in \mathbf{R}\}$ is the set of co-ordinates of the points on the plane. ■

Lemma 4.5.10: Let A, B, X and Y be sets.

(i) $(A \cup B) \times X = (A \times X) \cup (B \times X)$.
(ii) $(A \cap B) \times X = (A \times X) \cap (B \times X)$.
(iii) $(A - B) \times X = (A \times X) - (B \times X)$.
(iv) $(A \cap B) \times (X \cap Y) = (A \times X) \cap (B \times Y)$.
(v) $(A = \emptyset \vee B = \emptyset) \Leftrightarrow A \times B = \emptyset$.

Proof:

(i) (Bidirectional proof)

[**Note:** We shall use the second defining property of Cartesian product in our proofs.]

$(a,b) \in (A \cup B) \times X \Leftrightarrow a \in (A \cup B) \wedge b \in X$ (Definition of $\times$)
$\Leftrightarrow (a \in A \vee a \in B) \wedge b \in X$ (Definition of $\cup$)
$\Leftrightarrow b \in X \wedge (a \in A \vee a \in B)$ (E9)
$\Leftrightarrow (b \in X \wedge a \in A) \vee (b \in X \wedge a \in B)$ (E13)
$\Leftrightarrow (a \in A \wedge b \in X) \vee (a \in B \wedge b \in X)$ (E9)
$\Leftrightarrow (a,b) \in A \times X \vee (a,b) \in B \times X$ (Definition of $\times$)
$\Leftrightarrow (a,b) \in (A \times X) \cup (B \times X)$ (Definition of $\cup$)
$\therefore (\forall x)(x \in (A \cup B) \times X \Leftrightarrow x \in (A \times X) \cup (B \times X))$ (Gen)
Hence, $(A \cup B) \times X = (A \times X) \cup (B \times X)$ (Principle of Extension)

(ii) (Bidirectional proof)

$(a,b) \in (A \cap B) \times X \Leftrightarrow a \in (A \cap B) \wedge b \in X$ (Definition of $\times$)
$\Leftrightarrow (a \in A \wedge a \in B) \wedge b \in X$ (Definition of $\cap$)
$\Leftrightarrow (a \in A \wedge a \in B) \wedge (b \in X \wedge b \in X)$ (E3)
$\Leftrightarrow (a \in A \wedge b \in X) \wedge (a \in B \wedge b \in X)$ (E9, E11)
$\Leftrightarrow (a,b) \in (A \times X) \wedge (a,b) \in (B \times X)$ (Definition of $\times$)
$\Leftrightarrow (a,b) \in (A \times X) \cap (B \times X)$ (Definition of $\cap$)
$\therefore (\forall x)(x \in (A \cap B) \times X \Leftrightarrow x \in (A \times X) \cap (B \times X))$ (Gen)
Hence, $(A \cap B) \times X = (A \times X) \cap (B \times X)$ (Principle of Extension)

(iii) (Bidirectional proof)

$(a,b) \in (A - B) \times X \Leftrightarrow a \in (A - B) \wedge b \in X$ (Definition of $\times$)
$\Leftrightarrow (a \in A \wedge a \notin B) \wedge b \in X$ (Definition of $-$)

$\Leftrightarrow ((a \in A \wedge a \notin B) \wedge b \in X) \vee false$ (E6)

$\Leftrightarrow ((a \in A \wedge a \notin B) \wedge b \in X) \vee (a \in A \wedge false)$ (E7)

$\Leftrightarrow ((a \in A \wedge a \notin B) \wedge b \in X) \vee (a \in A \wedge (b \in X \wedge b \notin X))$ (E1)

$\Leftrightarrow (a \in A \wedge b \in X \wedge a \notin B) \vee (a \in A \wedge b \in X \wedge b \notin X)$ (E9, E11)

$\Leftrightarrow (a \in A \wedge b \in X) \wedge (a \notin B \vee b \notin X)$ (E13)

$\Leftrightarrow (a \in A \wedge b \in X) \wedge \sim (a \in B \wedge b \in X)$ (E16)

$\Leftrightarrow (a, b) \in (A \times X) \wedge \sim (a, b) \in (B \times X)$ (Definition of $\times$)

$\Leftrightarrow (a, b) \in (A \times X) - (B \times X)$ (Definition of $-$)

$\therefore (\forall x)(x \in (A - B) \times X \Leftrightarrow x \in (A \times X) - (B \times X))$ (Gen)

Hence, $(A - B) \times X = (A \times X) - (B \times X)$ (Principle of Extension)

(iv) (Bidirectional proof)

$(a, b) \in (A \cap B) \times (X \cap Y) \Leftrightarrow a \in (A \cap B) \wedge b \in (X \cap Y)$ (Definition of $\times$)

$\Leftrightarrow (a \in A \wedge a \in B) \wedge (b \in X \wedge b \in Y)$ (Definition of $\cap$)

$\Leftrightarrow (a \in A \wedge b \in X) \wedge (a \in B \wedge b \in Y)$ (E9, E11)

$\Leftrightarrow (a, b) \in (A \times X) \wedge (a, b) \in (B \times Y)$ (Definition of $\times$)

$\Leftrightarrow (a, b) \in (A \times X) \cap (B \times Y)$ (Definition of $\cap$)

$\therefore (\forall x)(x \in (A \cap B) \times (X \cap Y) \Leftrightarrow x \in (A \times X) \cap (B \times Y))$ (Gen)

Hence, $(A \cap B) \times (X \cap Y) = (A \times X) \cap (B \times Y)$ (Principle of Extension)

(v) (Bidirectional proof)

$\Rightarrow$) (Indirect Proof)

$A \times B \neq \emptyset \Rightarrow (\exists x)x \in A \times B$ (Lemma 3.5.8)

$\Rightarrow (a, b) \in A \times B$ (EI)

$\Rightarrow a \in A \wedge b \in B$ (Definition of $\times$)

$\Rightarrow a \in A$ and $b \in B$ (E9, I2)

$\Rightarrow (\exists x)x \in A$ and $(\exists x)x \in B$ (EQ)

$\Rightarrow A \neq \emptyset$ and $B \neq \emptyset$ (Lemma 3.5.8)

$\Rightarrow A \neq \emptyset \wedge B \neq \emptyset$ (I6)

$\Rightarrow \sim (A = \emptyset \vee B = \emptyset)$ (E17)

Therefore, $(A = \emptyset \vee B = \emptyset) \Rightarrow A \times B = \emptyset$.

$\Leftarrow$) (Indirect Proof)

$\sim (A = \emptyset \vee B = \emptyset) \Rightarrow A \neq \emptyset \wedge B \neq \emptyset$ (E17)

$\Rightarrow A \neq \emptyset$ and $B \neq \emptyset$ (E9, I2)

$\Rightarrow (\exists x)(x \in A)$ and $(\exists x)(x \in B)$ (Lemma 3.5.8)

$\Rightarrow a \in A$ and $b \in B$ (EI)

$\Rightarrow a \in A \wedge b \in B$ (I6)

$\Rightarrow (a, b) \in A \times B$ (Definition of $\times$)

$\Rightarrow (\exists x)(x \in A \times B)$ (EQ)

$\Rightarrow A \times B \neq \emptyset$ (Lemma 3.5.8)

Therefore, $A \times B = \emptyset \Rightarrow (A = \emptyset \vee B = \emptyset)$.

Hence, $(A = \emptyset \vee B = \emptyset) \Leftrightarrow A \times B = \emptyset$. ■

4.6 Operations on Collections of Sets

The union and intersection operators are defined for a pair of sets only. In this section, we shall generalize these operators to a collection of sets.

Definition: Let $\mathcal{C}$ be a collection of sets. The ***union of*** $\mathcal{C}$ is the set:

$$\bigcup_{X \in \mathcal{C}} X = \{x \mid (\exists X)(X \in \mathcal{C} \wedge x \in X)\} = \{x \mid x \in X \; for \; some \; X \in \mathcal{C}\}.$$

Remark: Recall that (see Section 3.3.2) the following notations are equivalent.

$(\exists x)(x \in X \wedge P(x))$, $(\exists x \in X)P(x)$, $\exists x \in X, P(x)$, and $P(x) \; for \; some \; x \in X$.

Example 4.6.6

1. Let $\mathcal{C} = \{\{a, e, i, o, u\}, \{a, b, c, d, e\}, \{p, q, r\}, \{1, 2, 3\}\}$.
   Then $\bigcup_{X \in \mathcal{C}} X = \{a, b, c, d, e, i, o, u, p, q, r, 1, 2, 3\}$.

2. When $\mathcal{C} = \emptyset$, $\bigcup_{X \in \mathcal{C}} X = \bigcup_{X \in \emptyset} X = \emptyset$

   Proof: (Proof by contradiction)

   Suppose $\bigcup_{X \in \emptyset} X \neq \emptyset$. Then $(\exists x) x \in \bigcup_{X \in \emptyset} X$ (Lemma 3.5.8)

   $\Rightarrow a \in \bigcup_{X \in \emptyset} X$ (EI)

   $\Rightarrow (\exists X)(X \in \emptyset \wedge a \in X)$ (Definition of $\bigcup_{X \in \emptyset} X$)

   $\Rightarrow (\exists X) X \in \emptyset \wedge (\exists X) a \in X$ (Section 2.8, Example 3)

   $\Rightarrow (\exists X) X \in \emptyset$ (I2)

   $\Rightarrow \emptyset \neq \emptyset$ (Lemma 3.5.8)

   By Lemma 3.2.1(i), $\emptyset = \emptyset \Rightarrow (\emptyset = \emptyset) \wedge (\emptyset \neq \emptyset)$ (I6)

   $\Rightarrow false$ (E1). (i.e. a contradiction)

   Hence, $\bigcup_{X \in \emptyset} X = \emptyset$.

3. When $\mathcal{C} = \{A\}$, $\bigcup_{X \in \mathcal{C}} X = \bigcup_{X \in \{A\}} X = A$

   Proof: (Left as an exercise).

4. When $\mathcal{C} = \{A, B\}$, $\bigcup_{X \in \mathcal{C}} X = \bigcup_{X \in \{A, B\}} X = A \cup B$

   Proof: (Left as an exercise).

Definition: Let $\mathcal{C}$ be a collection of sets such that $\mathcal{C} \neq \emptyset$. The ***intersection of*** $\mathcal{C}$ is the set:

$$\bigcap_{X \in \mathcal{C}} X = \{x \mid (\forall X)(X \in \mathcal{C} \Rightarrow x \in X)\} = \{x \mid x \in X \; for \; all \; X \in \mathcal{C}\}.$$

Remark:

1. Recall that (see Section 3.3) the following notations are equivalent:

   $(\forall x)(x \in X \Rightarrow P(x))$, $(\forall x \in X)P(x)$, $\forall x \in X, P(x)$, $P(x), \forall x \in X$, and $P(x) \; for \; all \; x \in X$.

2. In the above definition, $\mathcal{C}$ is required to be a *non-empty* set. This is because when $\mathcal{C} = \emptyset$, $\bigcap\limits_{X \in \emptyset} X$ is undefined. Formally, $\sim (\exists Y)\left(Y = \bigcap\limits_{X \in \emptyset} X\right)$. This is verified as follows.

Proof: (Proof by Contradiction) Suppose $(\exists Y)\left(Y = \bigcap\limits_{X \in \emptyset} X\right)$. Then $\mathbf{U} = \bigcap\limits_{X \in \emptyset} X$. (EI) . . . (I)

First, we want to prove that $(\forall x) x \in \mathbf{U}$ . . . . . (II)

(Proof by Contradiction) Suppose to the contrary that $\sim (\forall x) x \in \mathbf{U}$.

Then, $(\exists x) x \notin \mathbf{U}$ (FE7)

$\Rightarrow (\exists x) x \notin \bigcap\limits_{X \in \emptyset} X$ ((I), Sub$_=$)

$\Rightarrow a \notin \bigcap\limits_{X \in \emptyset} X$ (EI)

$\Rightarrow \sim (\forall X)(X \in \emptyset \Rightarrow a \in X)$ (Definition of $\bigcap$)

$\Rightarrow (\exists X) \sim (X \in \emptyset \Rightarrow a \in X)$ (FE7)

$\Rightarrow (\exists X) \sim (\sim X \in \emptyset \vee a \in X)$ (E18)

$\Rightarrow (\exists X)(\sim\sim X \in \emptyset \wedge a \notin X)$ (E17)

$\Rightarrow (\exists X)(X \in \emptyset \wedge a \notin X)$ (E15)

$\Rightarrow (\exists X)(X \in \emptyset) \wedge (\exists X) a \notin X$ (Section 2.8, Example 3)

$\Rightarrow (\exists X)(X \in \emptyset)$ (I2)

$\Rightarrow \emptyset \neq \emptyset$ (Lemma 3.5.8)

But $\emptyset = \emptyset$ (Lemma 3.2.1(i)). We thus have a contradiction!

It follows that proposition (II) holds.

Note that proposition (II) essentially states that $\mathbf{U} = \bigcap\limits_{X \in \emptyset} X$ is the *non-existing* absolute universal set. The rest of the proof is thus similar to that of the non-existence of the absolute universal set.

By the Principle of Specification, there exists a set $B = \{x \in \mathbf{U} \mid x \notin x\}$.

Now, either $B \in B$ or $B \notin B$.

(i) Suppose $B \in B$. Then B must satisfy the defining property of B which implies that $B \in \mathbf{U} \wedge B \notin B \Rightarrow B \notin B$. We thus have $B \in B \wedge B \notin B \Rightarrow false$, (i.e a contradiction).

(ii) Suppose $B \notin B$. Then from (I), by UI, we have: $B \in \mathbf{U}$. It follows that $B \in \mathbf{U} \wedge B \notin B$ (by I6) which implies that B satisfies the defining property of B. Hence, $B \in B$. We again have $B \in B \wedge B \notin B \Rightarrow false$, a contradiction!

Since in either case, a contradiction is derived, we thus conclude that the set $\bigcap\limits_{X \in \emptyset} X$ does not exist.

■

Example 4.6.7

1. Let $\mathcal{C} = \{\{a, c, f, h, k\}, \{a, b, c, h\}, \{a, c, h, k, m, n, p\}, \{b, c, g, h, n, r, x, y, z\}\}$.
   Then $\bigcap\limits_{X \in \mathcal{C}} X = \{c, h\}$.

2. When $\mathcal{C} = \{A\}$, $\bigcap\limits_{X \in \mathcal{C}} X = \bigcap\limits_{X \in \{A\}} X = A$
   Proof: (Left as an exercise).

3. When $C = \{A, B\}$, $\bigcap_{X \in C} X = \bigcap_{X \in \{A,B\}} X = A \cap B$.

Proof: (Left as an exercise). ■

4.7 Examples

1. Let X and Y be sets. Prove that:

(a) $X = (X - Y) \cup (X \cap Y)$.
(b) $(X - Y) \cap (X \cap Y) = \emptyset$.

Solution:

(a) $x \in X \Leftrightarrow x \in X \wedge true$ (E5)
$\Leftrightarrow x \in X \wedge (x \in Y \vee x \notin Y)$ (E2)
$\Leftrightarrow (x \in X \wedge x \in Y) \vee (x \in X \wedge x \notin Y)$ (E13)
$\Leftrightarrow (x \in X \cap Y) \vee (x \in X \wedge x \notin Y)$ (Definition of $\cap$)
$\Leftrightarrow (x \in X \cap Y) \vee (x \in X - Y)$ (Definition of $-$)
$\Leftrightarrow (x \in X - Y) \vee (x \in X \cap Y)$ (E10)
$\Leftrightarrow x \in (X - Y) \cup (X \cap Y)$ (Definition of $\cup$)
$\therefore (\forall x)(x \in X \Leftrightarrow x \in (X - Y) \cup (X \cap Y))$ (Gen)
Hence, $X = (X - Y) \cup (X \cap Y)$ (Principle of Extension)

(b) $x \in (X - Y) \cap (X \cap Y) \Leftrightarrow x \in (X - Y) \wedge x \in (X \cap Y)$ (Definition of $\cap$)
$\Leftrightarrow (x \in X \wedge x \notin Y) \wedge x \in (X \cap Y)$ (Definition of $-$)
$\Leftrightarrow (x \in X \wedge x \notin Y) \wedge (x \in X \wedge x \in Y)$ (Definition of $\cap$)
$\Leftrightarrow (x \in X \wedge x \in X) \wedge (x \in Y \wedge x \notin Y)$ (E9,E11)
$\Leftrightarrow (x \in X \wedge x \in X) \wedge false$ (E1)
$\Leftrightarrow false$ (E7)
$\Leftrightarrow x \in \emptyset$ (Lemma 3.5.7)
$\therefore (\forall x)(x \in (X - Y) \cap (X \cap Y) \Leftrightarrow x \in \emptyset)$ (Gen)
Hence, $(X - Y) \cap (X \cap Y) = \emptyset$ (Principle of Extension) ■

2. Prove that $A \cap B \subseteq (A \cap C) \cup (B \cap \overline{C})$.

Proof: We shall first prove that $x \in (A \cap B) \Rightarrow x \in (A \cap C) \cup (B \cap \overline{C})$.
(Direct Proof)
Let $x \in (A \cap B)$. Then $(x \in A \wedge x \in B)$ (Definition of $\cap$)
$\Rightarrow x \in A$ and $x \in B$ (I2,E9) . . . (I)
(Proof by cases)
Case (i): Suppose $x \in C$. . . . (II)
Then $(x \in A \wedge x \in C)$ ((I),(II),I6)
$\Rightarrow x \in (A \cap C)$ (Definition of $\cap$)
$\Rightarrow x \in (A \cap C) \vee x \in (B \cap \overline{C})$ (I1)
$\Rightarrow x \in (A \cap C) \cup (B \cap \overline{C})$ (Definition of $\cup$)
Case (ii): Suppose $x \notin C$.

Then $x \notin C \Rightarrow x \in \overline{C}$. (Lemma 4.3.5) . . . (III)

$\Rightarrow x \in B \wedge x \in \overline{C}$ ((I),(III),I6)

$\Rightarrow x \in (B \cap \overline{C})$ (Definition of $\cap$)

$\Rightarrow x \in (B \cap \overline{C}) \vee x \in (A \cap C)$ (I1)

$\Rightarrow x \in (A \cap C) \vee x \in (B \cap \overline{C})$ (E10)

$\Rightarrow x \in (A \cap C) \cup (B \cap \overline{C})$ (Definition of $\cup$)

Therefore, $x \in A \cap B \Rightarrow x \in (A \cap C) \cup (B \cap \overline{C})$

$\Rightarrow (\forall x)(x \in A \cap B \Rightarrow x \in (A \cap C) \cup (B \cap \overline{C}))$ (Gen)

$\Rightarrow A \cap B \subseteq (A \cap C) \cup (B \cap \overline{C})$. (Definition of $\subseteq$) ■

3. Prove that $(A \cup C) \cap (B \cup \overline{C}) \subseteq A \cup B$.

Proof: We shall first prove that $x \in (A \cup C) \cap (B \cup \overline{C}) \Rightarrow x \in A \cup B$.

(Direct Proof)

$x \in (A \cup C) \cap (B \cup \overline{C})$

$\Rightarrow x \in (A \cup C) \wedge x \in (B \cup \overline{C})$ (Definition of $\cap$)

$\Rightarrow (x \in A \vee x \in C) \wedge (x \in B \vee x \in \overline{C})$ (Definition of $\cup$)

$\Rightarrow (x \in A \vee x \in C)$ and $(x \in B \vee x \in \overline{C})$ (I2, E9,I2)

$(x \in A \vee x \in C) \Rightarrow (\sim x \notin A \vee x \in C)$ (E15)

$\Rightarrow (x \notin A \Rightarrow x \in C)$ (E18)

$(x \in B \vee x \in \overline{C}) \Rightarrow (x \in B \vee x \notin C)$ (Lemma 4.3.5)

$\Rightarrow (x \notin C \vee x \in B)$ (E10)

$\Rightarrow (x \in C \Rightarrow x \in B)$ (E18)

We thus have: $(x \notin A \Rightarrow x \in C)$ and $(x \in C \Rightarrow x \in B)$

$\Rightarrow (x \notin A \Rightarrow x \in B)$ (I5)

$\Rightarrow (\sim x \notin A \vee x \in B)$ (E18)

$\Rightarrow (x \in A \vee x \in B)$ (E15)

$\Rightarrow x \in A \cup B$ (Definition of $\cup$)

Hence, $x \in (A \cup C) \cap (B \cup \overline{C}) \Rightarrow x \in (A \cup B)$

$\Rightarrow (\forall x)(x \in (A \cup C) \cap (B \cup \overline{C}) \Rightarrow x \in (A \cup B))$ (Gen)

$\Rightarrow (A \cup C) \cap (B \cup \overline{C}) \subseteq A \cup B$. (Definition of $\subseteq$) ■

4. If $A \cup B = \mathbf{U}$ and $A \cap B = \emptyset$, then $B = \overline{A}$.

Proof: (Direct Proof) Suppose $A \cup B = \mathbf{U}$ and $A \cap B = \emptyset$. . . . (A)

To prove that $B = \overline{A}$, we shall first prove that $x \in B \Leftrightarrow x \in \overline{A}$

(Bidirection proof)

$\Rightarrow$) (Proof by contradiction)

Suppose $x \in B$ and $x \notin \overline{A}$.

Then $x \notin \overline{A} \Rightarrow x \in A$ (Lemma 4.3.5, E15)

So, $x \in A$ and $x \in B \Rightarrow x \in A \wedge x \in B$ (I6)

$\Rightarrow x \in (A \cap B)$ (Definition of $\cap$)

$\Rightarrow x \in \emptyset$ ((A),Sub$_=$)

$\Rightarrow false$ (Lemma 3.5.7)

Therefore, $x \in B \Rightarrow x \in \overline{A}$.

$\Leftarrow$) (Direct proof)

Suppose $x \in \overline{A}$. Then $x \notin A$ (Lemma 4.3.5) . . . (I)

By definition of **U**, $x \in \mathbf{U}$

$\Rightarrow x \in (A \cup B)$ ((A), Sub$_=$)

$\Rightarrow x \in A \vee x \in B$ (Definition of $\cup$)

$\Rightarrow \sim\sim x \in A \vee x \in B$ (E15)

$\Rightarrow (x \notin A \Rightarrow x \in B)$ (E18) . . . (II)

$\Rightarrow x \in B$ (I),(II), I3

Therefore, $x \in \overline{A} \Rightarrow x \in B$.

Hence, $(x \in B \Leftrightarrow x \in \overline{A})$ (E20)

$\Rightarrow (\forall x)(x \in B \Leftrightarrow x \in \overline{A})$ (Gen)

$\Rightarrow B = \overline{A}$ (Principle of Extension) ■

4.8 The Algebra of Sets

In this chapter and the last, we established lemmas and theorems that describe basic properties of $\subseteq$, $\cup, \cap, -$ and $\overline{\ }$, as well as their interrelationships. These lemmas and theorems are a part of what could be appropriately called the *algebra of sets*. In this section, we shall show that many of the set-theoretic problems we solved earlier can be redone using solely the algebra of sets.

First, we shall define expressions that involve set variables and set operators. These expressions are called ***set expression***s.

Definition:

(a) Every variable (such as A, B, X, etc.) and the constants **U** and $\emptyset$ is a set expression.

(b) If α and β are set expressions, then so are $\overline{\alpha}, (\alpha \cup \beta), (\alpha \cap \beta), (\alpha - \beta)$ and $(\alpha \times \beta)$.

(c) No expression other than those that can be defined by means of a finite number of applications of (a) or (b) is a set expression.

As with well-formed formulas, we may abbreviate a set expression by dropping some of the parenthese. To do so, we adopt the *left-associative* rule for the set operators: $\cup, \cap, -$ and $\times$.

1. Redo the proof of Corollary 4.2.2.1 using the algebra of sets.

Proof: First, we want to prove that $A \cap B \subseteq A$ using Theorem 4.2.2(v).

$(A \cap B) \cap A = A \cap (A \cap B)$ (Theorem 4.2.2(iii))

$= (A \cap A) \cap B$ (Theorem 4.2.2(iv))

$= A \cap B$ (Theorem 4.2.2(ii))

Therefore, $A \cap B \subseteq A$. (Theorem 4.2.2(v))

Next, we want to prove that $A \subseteq A \cup B$ using Theorem 4.1.1(v).

$A \cup (A \cup B) = (A \cup A) \cup B$ (Theorem 4.1.1(iv))

$= A \cup B$ (Theorem 4.1.1(ii))

Therefore, $A \subseteq A \cup B$. (Theorem 4.1.1(v)) ■

2. Redo the proof of Theorem 4.3.4(ii) and (iii) using the algebra of sets.

Solution:

Proof of (ii): $A - (A - B) = A - (A \cap \overline{B})$ (Theorem 4.3.7(v))

$= A \cap \overline{(A \cap \overline{B})}$ (Theorem 4.3.7(v))

$= A \cap \left(\overline{A} \cup \overline{\overline{B}}\right)$ (Theorem 4.3.6(ii))

$= A \cap (\overline{A} \cup B)$ (Theorem 4.3.7(i))

$= (A \cap \overline{A}) \cup (A \cap B)$ (Theorem 4.2.3(ii))

$= \emptyset \cup (A \cap B)$ (Theorem 4.3.7(iii))

$= (A \cap B) \cup \emptyset$ (Theorem 4.1.1(iii))

$= (A \cap B)$ (Theorem 4.1.1(i))

Proof of (iii): $(A \cap B) - (A \cap C) = (A \cap B) \cap \overline{(A \cap C)}$ (Theorem 4.3.7(v))

$= (A \cap B) \cap (\overline{A} \cup \overline{C})$ (Theorem 4.3.6(ii))

$= ((A \cap B) \cap \overline{A}) \cup ((A \cap B) \cap \overline{C})$ (Theorem 4.2.3(ii))

$= ((A \cap \overline{A}) \cap B) \cup ((A \cap B) \cap \overline{C})$ (Theorem 4.2.2(iii),(iv))

$= (\emptyset \cap B) \cup ((A \cap B) \cap \overline{C})$ (Theorem 4.3.7(iii))

$= (B \cap \emptyset) \cup ((A \cap B) \cap \overline{C})$ (Theorem 4.2.2(iii))

$= \emptyset \cup ((A \cap B) \cap \overline{C})$ (Theorem 4.2.2(i))

$= ((A \cap B) \cap \overline{C}) \cup \emptyset$ (Theorem 4.1.1(iii))

$= ((A \cap B) \cap \overline{C})$ (Theorem 4.1.1(i))

$= (A \cap (B \cap \overline{C}))$ (Theorem 4.2.2(v))

$= A \cap (B - C)$ (Theorem 4.3.7(v)) ■

3. Redo the proof of $\overline{\mathbf{U}} = \emptyset$ by using the algebra of sets.

Solution: $\emptyset = \emptyset$ (Theorem 3.2.1(i))

$\Rightarrow \overline{\overline{\emptyset}} = \emptyset$ (Theorem 4.3.7(i), $\text{Sub}_=$)

$\Rightarrow \overline{\mathbf{U}} = \emptyset$ (Theorem 4.3.7 (ii) first part, $\text{Sub}_=$) ■

4. Redo the proof of $A \cup \overline{A} = \mathbf{U}$ by using the algebra of sets.

Solution: $\overline{A \cap \overline{A}} = \overline{A \cap \overline{A}}$ (Theorem 3.2.1(i))

$\Rightarrow \overline{A \cap \overline{A}} = \overline{\emptyset}$ (Theorem 4.3.7(iii) first part, $\text{Sub}_=$)

$\Rightarrow \overline{A} \cup \overline{\overline{A}} = \overline{\emptyset}$ (Lemma 4.3.6(ii))

$\Rightarrow \overline{A} \cup A = \overline{\emptyset}$ (Theorem 4.3.7(i))

$\Rightarrow \overline{A} \cup A = \mathbf{U}$ (Theorem 4.3.7(ii))

$\Rightarrow A \cup \overline{A} = \mathbf{U}$ (Lemma 4.1.1(ii)) ■

5. Redo Section 4.7, Example 1 using the algebra of sets.

Solution:

(a) $(X - Y) \cup (X \cap Y) = (X \cap \overline{Y}) \cup (X \cap Y)$ (Theorem 4.3.7(v))

$= X \cap (\overline{Y} \cup Y)$ (Theorem 4.2.3(ii))

$= X \cap (Y \cup \overline{Y})$ (Theorem 4.1.1(iii))

$= X \cap \mathbf{U}$ (Theorem 4.3.7(iii))

$= X$ ($X \subseteq \mathbf{U}$ and Theorem 4.2.2(v))

(b) $(X - Y) \cap (X \cap Y) = (X \cap \overline{Y}) \cap (X \cap Y)$ (Theorem 4.3.7(v))

$= (X \cap X) \cap (Y \cap \overline{Y})$ (Theorem 4.2.2(iii),(iv))

$= (X \cap X) \cap \emptyset$ (Theorem 4.3.7(iii))

$= \emptyset$ (Theorem 4.2.2(i)) ■

6. Redo Section 4.7, Example 2 using the algebra of sets.

Solution: To prove that $A \cap B \subseteq (A \cap C) \cup (B \cap \overline{C})$, we shall prove that $(A \cap B) - ((A \cap C) \cup (B \cap \overline{C})) = \emptyset$.

First,. we shall prove the following lemmas.

Lemma A: $(X \cap Y) \cup X = X$

Proof: $X \cap Y \subseteq X$ (Corollary 4.2.2.1) $\cdots\cdots$ (I)

$\Rightarrow (X \cap Y) \cup X = X$ ((I), and Theorem 4.1.1(v)) ■

$(A \cap B) - ((A \cap C) \cup (B \cap \overline{C}))$

$= (A \cap B) \cap \overline{(A \cap C) \cup (B \cap \overline{C})}$ (Theorem 4.3.7(v))

$= (A \cap B) \cap \overline{(((A \cap C) \cap B) \cup (A \cap C)) \cup (B \cap \overline{C})}$ (Lemma A, $X = (A \cap C)$, $Y = B$)

$= (A \cap B) \cap \overline{(((A \cap C) \cap B) \cup (A \cap C)) \cup (((B \cap \overline{C}) \cap A) \cup (B \cap \overline{C}))}$ (Lemma A, $X = (B \cap \overline{C})$, $Y = A$)

$= (A \cap B) \cap \overline{((A \cap C) \cup (B \cap \overline{C})) \cup ((A \cap C \cap B) \cup (B \cap \overline{C} \cap A))}$ (Theorem 4.1.1(iii),(iv))

$= (A \cap B) \cap \overline{((A \cap C) \cup (B \cap \overline{C})) \cup (((A \cap B) \cap C) \cup ((A \cap B) \cap \overline{C}))}$ (Theorem 4.2.2(iii),(iv))

$= (A \cap B) \cap \overline{((A \cap C) \cup (B \cap \overline{C})) \cup ((A \cap B) \cap (C \cup \overline{C}))}$ (Theorem 4.2.3(ii))

$= (A \cap B) \cap \overline{((A \cap C) \cup (B \cap \overline{C})) \cup ((A \cap B) \cap \mathbf{U})}$ (Theorem 4.3.7(iii))

$= (A \cap B) \cap \overline{((A \cap C) \cup (B \cap \overline{C})) \cup ((A \cap B))}$ (Definition of $\mathbf{U}$; Theorem 4.2.2(v))

$= (A \cap B) \cap \left(\overline{((A \cap C) \cup (B \cap \overline{C}))} \cap \overline{(A \cap B)}\right)$ (Theorem 4.3.6(i))

$= \left((A \cap B) \cap \overline{(A \cap B)}\right) \cap \overline{(A \cap C) \cup (B \cap \overline{C})}$ (Theorem 4.2.2(iii),(iv))

$= \emptyset \cap \overline{(A \cap C) \cup (B \cap \overline{C})}$ (Theorem 4.3.7(iii))

$= \overline{(A \cap C) \cup (B \cap \overline{C})} \cap \emptyset$ (Theorem 4.2.2(iii))

$= \emptyset$ (Theorem 4.2.2(i))

Hence, by Theorem 4.3.4(i), $A \cap B \subseteq (A \cap C) \cup (B \cap \overline{C})$. ■

7. Redo Section 4.7, Example 3 using the algebra of sets.

Solution: $\overline{A} \cap \overline{B} \subseteq (\overline{A} \cap \overline{C}) \cup (\overline{B} \cap \overline{\overline{C}})$ (Example 2 above)

$\Rightarrow \overline{(\overline{A} \cap \overline{C}) \cup (\overline{B} \cap \overline{\overline{C}})} \subseteq \overline{\overline{A} \cap \overline{B}}$ (Theorem4.3.7(iv))

$\Rightarrow \overline{(\overline{A} \cap \overline{C})} \cap \overline{(\overline{B} \cap \overline{\overline{C}})} \subseteq \overline{\overline{A}} \cup \overline{\overline{B}}$ (Theorem4.3.6(i) and (ii))

$\Rightarrow (\overline{\overline{A}} \cup \overline{\overline{C}}) \cap \overline{(\overline{B} \cap \overline{\overline{C}})} \subseteq \overline{\overline{A}} \cup \overline{\overline{B}}$ (Theorem4.3.6(ii))

$\Rightarrow (A \cup C) \cap \overline{(\overline{B} \cap C)} \subseteq A \cup B$ (Theorem4.3.7(i))

$\Rightarrow (A \cup C) \cap \left(\overline{\overline{B}} \cup \overline{C}\right) \subseteq A \cup B$ (Theorem4.3.6(ii))

$\Rightarrow (A \cup C) \cap (B \cup \overline{C}) \subseteq A \cup B$ (Theorem4.3.7(i)) ■

8. Redo Section 4.7, Example 4 using the algebra of sets.

Solution: Suppose $A \cup B = \mathbf{U}$ and $A \cap B = \emptyset$ .... (I)

Then $\overline{A \cup B} = \overline{A \cup B}$ (Lemma 3.2.1(i))

$\Rightarrow \overline{A \cup B} = \overline{\mathbf{U}}$ ((I), $\text{Sub}_=$)

$\Rightarrow \overline{A} \cap \overline{B} = \overline{\mathbf{U}}$ (Theorem 4.3.6(i))

$\Rightarrow \overline{A} \cap \overline{B} = \emptyset$ (Theorem 4.3.7(ii))

$\Rightarrow \overline{A} - B = \emptyset$ (Theorem 4.3.7(v))

$\Rightarrow \overline{A} \subseteq B$ (Theorem 4.3.4(i))

$A \cap B = \emptyset$

$\Rightarrow B \cap A = \emptyset$ (Theorem 4.2.2(iii))

$\Rightarrow B \cap \overline{\overline{A}} = \emptyset$ (Theorem 4.3.7(i))

$\Rightarrow B - \overline{A} = \emptyset$ (Theorem 4.3.7(v))

$\Rightarrow B \subseteq \overline{A}$ (Theorem 4.3.4(i))

We thus have: $B \subseteq \overline{A}$ and $\overline{A} \subseteq B$

$\Rightarrow B = \overline{A}$ (Lemma 3.2.3(ii)) ■

4.9 Exercises

1. Suppose x is a set such that $x \notin x$. Of which of the following sets is x a member, x a subset, x neither a member nor a subset?

   (a) $\{\{x\}, y\}$
   (b) x
   (c) $\emptyset \cap x$
   (d) $\{x\} - \{\{x\}\}$
   (e) $\{x\} \cup x$
   (f) $\{x\} \cup \{\emptyset\}$

2. Let A, B, C, D be sets such that $A = \emptyset \cup \{\emptyset\}$, $B = A \cup \{A\}$, $C = B \cup \{B\}$ and $D = C \cup \{C\}$.

   (a) Express the set D using only the symbols $\{$, $\}$, $\emptyset$ and ,.
   (b) Determine which of the following is true:

   $$A \in B; \quad A \cap B = \emptyset; \quad (\emptyset \cap B) \in C; \quad B \subseteq C; \quad C \cup D = D$$

3. Prove that if $A \subseteq C$ and $B \subseteq D$, then $A \cup B \subseteq C \cup D$.
4. Prove that $(A \cap B) \cup C = A \cap (B \cup C) \Leftrightarrow C \subseteq A$.
5. Use only the theorems in this chapter to prove the following:

   (a) $\overline{(\overline{A} \cap B) \cap (A \cup \overline{B}) \cap (A \cup C)} = (A \cup \overline{B}) \cup (\overline{A} \cap (B \cup \overline{C}))$
   (b) $(\overline{A} \cap B \cap C) \cup (\overline{A} \cap C \cap A) \cup (\overline{A} \cap B \cap C \cap \overline{D \cap E}) = \overline{A} \cap B \cap C$
   (c) $(\overline{A} \cap B \cap \overline{C} \cap D) \cup (A \cap \overline{C}) \cup (\overline{B} \cap \overline{C}) \cup (\overline{C} \cap \overline{D}) = \overline{C}$
   (d) $(A \cap C) \cup (B \cap \overline{C}) \cup (A \cap B) = (A \cap C) \cup (B \cap \overline{C})$

6. (a) Let $X \cup Y = X$ for any set X. Prove that $Y = \emptyset$.
   (b) Let $X \cap Y = X$ for any set X. Prove that $Y = \mathbf{U}$.
7. Prove that $A \cup B = A \cap B$ iff $A = B$
8. Prove that
   (a) $A \cap (B - A) = \emptyset$.
   (b) $A \cup (B - A) = A \cup B$.
   (c) $(A \cup B) - C = (A - C) \cup (B - C)$.
   (d) $(A - B) \cup (B - A) = (A \cup B) - (A \cap B)$.
   (e) $(A - B) - C = (A - C) - (B - C)$.
9. Let A, B, C be sets. Prove that:
   (a) $A \subseteq B$ iff $A \cap \overline{B} = \emptyset$.
   (b) $A \subseteq B$ iff $\overline{A} \cup B = \mathbf{U}$.
   (c) $A \subseteq B$ iff $(A \cap \overline{B}) \subseteq \overline{A}$.
   (d) $A \subseteq B$ iff $(A \cap \overline{B}) \subseteq B$.
10. Let A, B and C be sets. Prove that $A \cup B \cup C = (A - B) \cup (B - C) \cup (C - A) \cup (A \cap B \cap C)$.
11. Prove or disprove the following:
   (a) $(A \times B) \cup (C \times D) = (A \cup C) \times (B \cup D)$.
   (b) $\mathcal{P}(A \cup B) = \mathcal{P}(A) \cup \mathcal{P}(B)$.
   (c) $\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B)$.
   (d) $\mathcal{P}(A - B) = \mathcal{P}(A) - \mathcal{P}(B)$.
   (e) $\mathcal{P}(A \times B) = \mathcal{P}(A) \times \mathcal{P}(B)$.
12. Prove that if $A \subseteq B$, then there exists a set C such that $(A \cup C) \subseteq (B \cup C)$ and $(A \cap C) \subseteq (B \cap C)$.
13. Prove that if $X \times Y = X \times Z$ and $X \neq \emptyset$, then $Y = Z$.
14. Prove that:
   (a) $\bigcup_{X \in \{A\}} X = A$
   (b) $\bigcup_{X \in \{A,B\}} X = A \cup B$
   (c) $\bigcap_{X \in \{A\}} X = A$
   (d) $\bigcap_{X \in \{A,B\}} X = A \cap B$
15. A *set equation* in one variable X is an equation of the form $\alpha(X) = \beta(X)$, where X is an unknown set and $\alpha(X)$ and $\beta(X)$ are set expressions formed using $\cup, \cap, -$ and $\overline{\ }$ on X and some fixed sets $A_i, 1 \leq i \leq n$. The following procedure determines a condition under which the equation has a solution for X, and all solutions if the condition is satisfied.

Procedure:
begin

1. Given $\alpha(X) = \beta(X)$. Construct $\alpha(X) \oplus \beta(X) = \emptyset$. (note: $A \oplus B = (A \cap \overline{B}) \cup (B \cap \overline{A})$)
2. Convert $\alpha(X) \oplus \beta(X) = \emptyset$ into an equation of the form $(\alpha \cap X) \cup (\beta \cap \overline{X}) = \emptyset$ such that the set expressions α and β are free of X, as follows:
   (a) repeatedly apply Theorem 4.3.7(v) to replace subexpressions of the form $(A - B)$ to $(A \cap \overline{B})$ until no such expression exists;
   (b) repeatedly apply Theorem 4.3.6 (DeMorgan's Theorem) to subexpressions of the form $\overline{A}$ until no such application can be applied;
   (c) repeatedly apply Theorem 4.3.7(i) to subexpressions of the form $\overline{\overline{A}}$ (note: $A = X$ or $A \in \{A_i \mid 1 \leq i \leq n\}$) until no such subexpression exists;

(d) repeatedly apply Theorem 4.2.3 (the distributed laws) to expand subexpressions of the form $A \cap (B \cup C)$ or $(B \cup C) \cap A$ until an expression of the form $\bigcup_{i=1}^{m} \left(A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_{k_i}} \right)$ is obtained, where each A_{i_j} is an *individual set* or the *absolute complement of an individual set*;

(e) For each $i, 1 \leq i \leq m$, replace $\left(A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_{k_i}} \right)$ with $\left(A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_{k_i}} \cap X \right) \cup \left(A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_{k_i}} \cap \overline{X} \right)$ if the subexpression contains neither X nor $\overline{X}$;

(f) Group all the subexpressions formed in Step (*e*) that contain X together and apply Theorem 4.2.3 (the distributed laws) to obtain an expression of the form $\alpha \cap X$, where α is free of X;
Group all the subexpressions formed in Step (*e*) that contain $\overline{X}$ together and apply Theorem 4.2.3 (the distributed laws) to obtain an expression of the form $\beta \cap \overline{X}$, where β is free of $\overline{X}$;
[Note: $\alpha(X) \oplus \beta(X) = \emptyset$ has been transformed into $(\alpha \cap X) \cup (\beta \cap \overline{X}) = \emptyset$.]

3. Declare that the given equation $\alpha(X) = \beta(X)$ has a solution if and only if the condition $\beta \subseteq \overline{\alpha}$ holds true and that any X such that $\beta \subseteq X \subseteq \overline{\alpha}$ is a solution.

end.

(i) Prove that $A \cup B = \emptyset \Leftrightarrow A = \emptyset \wedge B = \emptyset$.

(ii) Prove that the procedure correctly procedures a sufficient and necessary condition and the solution set.

(iii) Use the procedure to determine a sufficient and necessary condition for each of the following equations to have a solution:

a. $X \cup C = D$

b. $(A \cap X) \cup C = A \cap (X \cup C)$

c. $X \cup Y = X$

d. $A \cup X = A \cap X$

(iv) Prove that the equation $(A \cap X) \cup (B \cap \overline{X}) = (C \cap X) \cup (D \cap \overline{X})$ has a solution iff $B \ominus D \subseteq \overline{A \oplus C}$ and determine all the solutions.

(v) Prove that an alternative form for the solutions is $X = (\beta \cup Y) \cap \overline{\alpha}$, for some set Y.